

An Autonomous CZT Module for X-ray Diffraction Imaging

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Abstract—We present the development of a CZT-based detection module dedicated to X-ray diffraction imaging. This kind of application requires a good energy and spatial resolution in order to resolve Bragg peaks.

In a first part, we present the detector configuration used and dimensioning constraints. As the input energy range is comprised between 20 and 150 keV, we use 5 mm thick high resistivity CZT crystals. The 660 mm² detection area is segmented on both sides into 192 anodes and 12 cathodes. Signals from both sides are read jointly in order to perform multiparametric event corrections (depth of interaction, charge sharing, induction sharing). In order to be integrated easily inside an X-ray imaging system, the system has been conceived to be completely autonomous: it is powered by a single 12V supply and is interfaced with the external system by Ethernet for communication and RS485 for synchronization.

In a second part, we describe the system readout architecture and then the implementation of the data processing. An FPGA circuit embeds a digital processing chain that carries out readout ASIC interfacing and advanced multiparametric data corrections. Gain, offset but also depth of interaction and charge sharing are corrected on the flow. Incoming events from different channels are clustered together by comparing their location and time of occurrence. The FPGA also embeds a processor running an operating system that controls the system, carries out all calibrations, automated tests and acquisitions.

Eventually, we show the results obtained and demonstrate the relative influence of depth of interaction and charge sharing. Homogeneity of detector behavior is also discussed and the reproducibility of the performance between modules is presented. The average energy resolution at 25°C is 2.4 % FWHM at 122 keV and 3.8 % FWHM at 60keV and the average efficiency is 73 %.

I. INTRODUCTION

X-RAY diffraction (XRD) has been proven to be an effective technique for hold baggage screening in airport. X-ray diffraction imaging (XDi) is a new imaging architecture proposed by Morpho Detection in order to considerably improve throughput of diffraction baggage scanners and to enable XRD screening of cabin baggages [1]. The proposed XDi scanner design is based on the use of room-temperature operated CZT detectors in replacement of cooled Germanium. CZT detectors allows to build compact and modular detection units with an energy resolution compatible with diffraction pattern identification [2], [3].

The detection module, called IMADIF (by combination of IMaging and DIffraction) was designed to be integrated

inside the XDi cabin baggage screener and had to fulfill the following requirements: easy installation and replacement, simple interfacing and integration and good and stable spectral performance in the 20-150 keV range.

As the system has to be integrated in a complex system, ease of use and autonomy were key design criteria. The whole system is a relatively compact device (see Fig. 1). The module features a high level software interface hiding complexities of detector calibration and multi-parametric corrections behind simple commands carried by TCP network packets.



Fig. 1. External view of the detection module. It measures 170 mm × 75 mm × 78 mm for a power dissipation of 3.5 Watts.

II. DETECTOR DESIGN

A. Motivation and requirements

When diffraction signatures are collected, one probes electron distribution. The cross-section with a single electron is given by the Thomson factor [4]:

$$\frac{d\sigma_T}{d\Omega}(\theta) = r_e^2 \cdot \frac{1 + \cos^2 \theta}{2} \quad (1)$$

where $r_e = \frac{\mu_0 e^2}{4\pi m_e}$ is the classical electron radius. The cross-section with a given material depends on the electron distribution through the form factor $F(\Delta p)$,

$$\frac{d\sigma}{d\Omega}(E, \theta) = \frac{d\sigma_T}{d\Omega}(\theta) F^2(\Delta p) \quad (2)$$

Δp being the momentum transfer between incident and scattered photon, given by Bragg law:

$$\Delta p = 2 \frac{E}{c} \sin \frac{\theta}{2} \quad (3)$$

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In conclusion, to be able to accurately probe the material structure, indicated by the form factor, it is essential to combine a good energy and angular resolution. Moreover, both have to be consistent, as the final resolution on momentum transfer results from the convolution of both uncertainties. Angular resolution results from a trade-off with sensitivity through collimation opening and thus it is limited to what is needed to discriminate threats from innocent material. As the spatial resolution is mainly needed to have a small angular uncertainty, spatial resolution is more important in one direction (the angular deviation direction defined by the secondary collimation). Detector anode side is thus segmented in rectangular shaped electrodes.

As angular resolution itself is limited by sensitivity/resolution trade-off, the high energy resolution of a HPGe detector is not needed for a high throughput system and CZT detector resolution (few percents) is sufficient. Count rate requirements are also relatively modest due to the low flux of diffracted photon compared with the incident X-ray flux. High flux grade detector and electronics are thus not needed too and a conventional spectrometric chain is well adapted.

B. Detector configuration

As efficiency is an important criterion and because most incident X-rays are in the 10-100keV range, 5 mm thick detector have been used. For such thickness, CZT high resistivity material are interesting and provide a good spectral behavior. We used a anode/cathode readout scheme to be able to correct depth of interaction effects. To define the anode size, we first carried out simulations using Penelope [4] and our detector simulation tools [6]. Spectra of various configuration have been simulated (Fig. 2), including effect of depth of interaction biparametric correction to define the most interesting configurations.

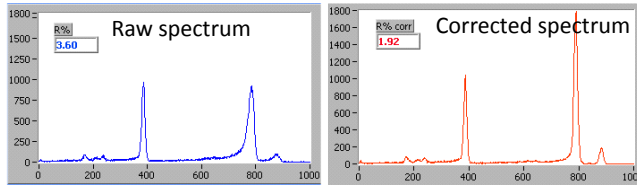


Fig. 2. Simulation of a Cobalt 57 and Americium 241 spectrum before and after biparametric correction. 5 mm thick CZT biased at 700V, 4.7 mm² electrode. Resolution before correction is 3.6 % and 1.9 % after (for an idealized detector).

Afterwards, we used a CZT test device to experimentally compare performance of various configuration (see Fig. 3).

Spectral performance of CZT detector mainly depends on two factors: readout noise [5] and detector response non-uniformity, especially variability due to depth of interaction effect and charge sharing. Large electrodes suffer from peak tailing and high electric noise, while small electrodes suffer from charge sharing and are more challenging in terms of readout integration and power dissipation. We have chosen to use 3.4 mm² electrodes as a reasonable trade-off. Observed noise

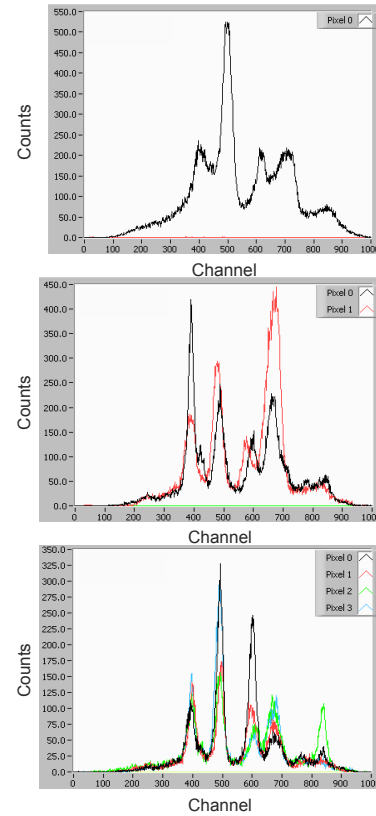


Fig. 3. Diffraction spectra of the same material obtained on a detection line subdivided in 1, 2 or 4 anodes. Corresponding anode area are 9.3, 4.7 and 2.3 mm².

on these electrodes (Fig. 4) is typically between 170 and 220 rms electrons at 500 V bias; the variability essentially comes from leakage current variability (0.2 to 1.5 nA). Additional stray capacitance due to CZT and PCB is around 2.5 pF and dielectric noise contributes to Equivalent Noise charge (ENC) for 60 rms electrons. Shaping time used is short (0.7 μ s) to cope with shot noise contribution.

The 660 mm² detection area is thus divided in 192 anodes to limit readout noise to an acceptable value and have a good pixel effect. We decided to use cathode readout with a 12 cathode segmentation to properly estimate depth of interaction. We also decided to handle multi-channel signal readout to correct charge-sharing effects. The system is operated at 700 V bias in order to balance shot noise and charge sharing degradation effects.

III. SYSTEM ELECTRONICS

A. Readout integrated circuits

The readout system is relatively complex as it uses two different kinds of ASIC: one for anode readout and one for cathode readout. These two ASICs were on-the-shelf and were designed prior the design of the detection module itself.

The anode ASIC, called IDeFX-HD [8] is a low-noise low-power circuit designed to read a high number of small area electrodes with low stray capacitance and low leakage current

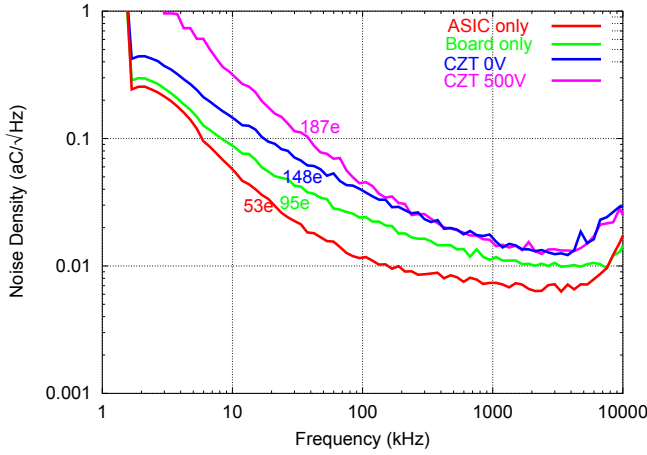


Fig. 4. Observed charge noise spectral density in four configurations. Red: with ASIC only and a internal 20pA source (ENC=53 rms electrons). Green: with ASIC and detector hybridization board (ENC=95 rms electrons). Blue: with unbiased detector (ENC=148 rms electrons). Purple: with 500V biased detector (ENC=187 rms electrons).

(typically less than 2 pF and 1 nA). It dissipates 1 mW per channel.

The cathode ASIC, called ALIGASPECT is a transient digitizing ASIC adapted to large electrodes, with typically 10 pF capacitance and up to 20 nA leakage currents. The current driven through input transistor is notably higher and it dissipates 6 mW per channel.

The total power dissipation for all ASICs is less than 300 mW.

B. Digital electronics

The system is driven by a single FPGA that handle all system control, data processing and interfacing.

We use a multi-electrode readout scheme [7] to enable advanced data corrections. Multiples anodes and cathodes signals can be read for a single interaction event. Depth of interaction is computed from cathode over anode ratio while charge sharing is computed by comparing charge collected between anodes. The hardware data processing pipeline (Fig. 5) carries out real time corrections:

- it communicates with ASICs,
- performs gain and offset corrections,
- sort and cluster coincident detected events,
- corrects amplitudes from depth of interaction and charge sharing to compensate for CIE non-uniformities,
- send photon data in Ethernet UDP packets.

An embedded processor (Altera NIOS II) allows to perform complex test and calibration sequences, also handles system monitoring and configuration. The module is then easy to manage and to integrate into the full system.

IV. RESULTS

Thanks to the low noise of readout ASIC, the good quality of CZT crystals, and the applied correction, performance is good and reproducible at room temperature (modules are regulated to 25 °C). Figure 6 shows two example spectra as

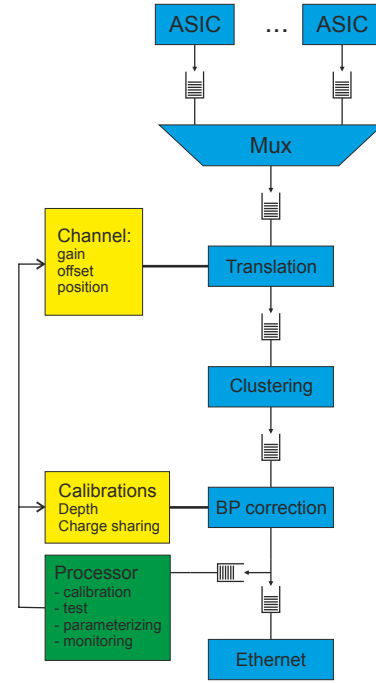


Fig. 5. Event processing pipeline

measured from the output of the module. CIE corrections are applied on the event flow and are then transparent for the end-user.

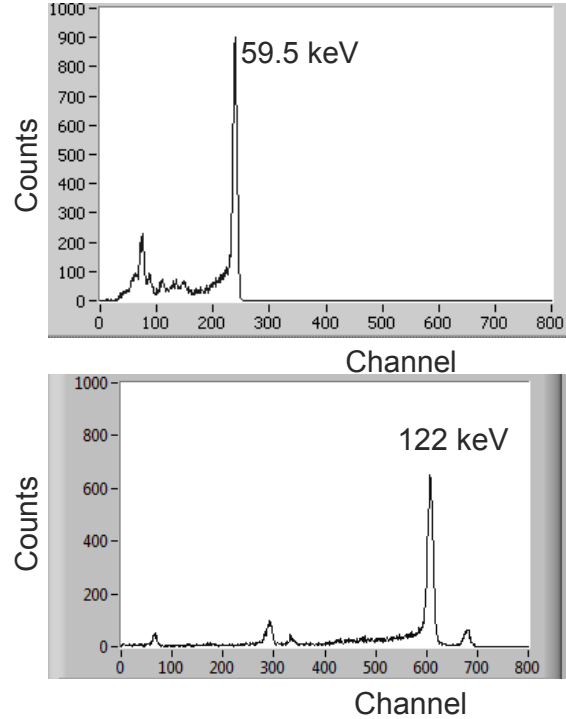


Fig. 6. Spectral performance of the module (on one good performing anode) with Americium-241 (top) and Cobalt-57 (bottom) sources and full area irradiation. Energy resolution are here respectively 3.2 % and 2.3 %

TABLE I
STATISTICS ON 83 DETECTORS

	Average	Std deviation
Dead pixels	1.7	2.8
Resolution (pulser)	2.1 keV	0.1 keV
Efficiency (60keV)	73 %	4 %
Resolution (60keV)	3.8 %	0.2 %
Resolution (122keV)	2.4 %	0.2 %

Statistically speaking (Table I), a quite stable performance has been observed on a large amount of detectors (83 detectors of 192 anodes). In average, almost two dead pixels (badly connected or noisy) are present on each detector device. It can be tolerated in a certain limit but it impacts global efficiency. Room temperature energy resolution averaged on all module are comprised between 3.4 and 4.3 % at 60 keV and between 1.9 and 2.7 % at 122 keV. Efficiency is in general quite reproducible between detectors (average between 68 and 75 % at 60 keV) but less homogeneous inside a given detector. Note that efficiency has been measured in a 10 % wide window (6 keV) centered on 60 keV.

V. CONCLUSION

We have developed a small but autonomous detection system to enable the production of innovative baggage scanners based on the new technology of X-ray diffraction imaging. Data processing, calibration and test functions are fully integrated inside the module, making it easy to use and manage. These modules perform correctly, with a good stability and reproducibility. They were integrated inside a scanner prototype that has demonstrated the feasibility of a XRD based baggage screener operating at high throughput (up to 600 bags per hour).

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